

Wazuh Upgrade & First Agent Session

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Session Summary

Task	Status
Snapshot taken before upgrade (pre-wazuh-upgrade-v2)	Done
Wazuh indexer upgraded to 4.14.6	Done
Wazuh manager upgraded to 4.14.6	Done
Wazuh dashboard upgraded to 4.14.6	Done
Cert ownership fixed proactively (wazuh-dashboard:wazuh-dashboard)	Done
Dashboard cert regenerated (ENOENT on dashboard-key.pem)	Done
Indexer certs updated (node-1.pem, node-1-key.pem, root-ca.pem)	Done
Passwords reset via wazuh-passwords-tool --change-all	Done
IPv6/IPv4 localhost fix (localhost -> 127.0.0.1 in opensearch_dashboards.yml)	Done
Vaultwarden agent registered and active (v4.14.6)	Done
CIS Benchmark scan completed — 39% score	Done
7 Critical CVEs identified in Vaultwarden packages	Done

1. Upgrade Process

1.1 Upgrade Order

Components must be upgraded in this order — each depends on the one before it:

1. Wazuh Indexer (OpenSearch – data store)
2. Wazuh Manager (analysis engine)
3. Wazuh Dashboard (web UI)

Why order matters: The manager writes data in formats the indexer must understand. The dashboard reads from both. Upgrading out of order risks data format mismatches or failed connections.

1.2 Upgrade Commands

```
apt install wazuh-indexer -y
apt install wazuh-manager -y
apt install wazuh-dashboard -y
```

Using apt install on an already-installed package upgrades it to the latest available version. Using apt install instead of apt upgrade gives control over which components upgrade and in what order.

2. Troubleshooting

2.1 Dashboard Cert Missing (ENOENT)

```
Error: ENOENT: no such file or directory, open '/etc/wazuh-dashboard/certs/dashboard-key.pem'
```

Cause: Dashboard upgrade wiped the certs directory. Fix — regenerate certs and copy with correct filenames:

```
curl -sO https://packages.wazuh.com/4.14/config.yml
# Edit config.yml — set all IPs to 192.168.10.71
rm /root/wazuh-install-files.tar
bash wazuh-install.sh -g
tar -xvf wazuh-install-files.tar
cp wazuh-install-files/dashboard.pem /etc/wazuh-dashboard/certs/dashboard.pem
cp wazuh-install-files/dashboard-key.pem /etc/wazuh-dashboard/certs/dashboard-key.pem
cp wazuh-install-files/root-ca.pem /etc/wazuh-dashboard/certs/
```

2.2 Cert Ownership Fix (EACCES)

```
Error: EACCES: permission denied, open '/etc/wazuh-dashboard/certs/dashboard-key.pem'
```

Cause: Copied certs owned by root but wazuh-dashboard process runs as wazuh-dashboard user. Fix proactively after every cert copy:

```
chown -R wazuh-dashboard:wazuh-dashboard /etc/wazuh-dashboard/certs/
chmod 500 /etc/wazuh-dashboard/certs/
chmod 400 /etc/wazuh-dashboard/certs/*
```

2.3 Indexer Cert Mismatch

Dashboard connected to indexer but got connection refused. New dashboard certs didn't match indexer's certs. Also need to update indexer certs:

```
cp wazuh-install-files/node-1.pem /etc/wazuh-indexer/certs/wazuh-indexer.pem
cp wazuh-install-files/node-1-key.pem /etc/wazuh-indexer/certs/wazuh-indexer-key.pem
cp wazuh-install-files/root-ca.pem /etc/wazuh-indexer/certs/root-ca.pem
chown -R wazuh-indexer:wazuh-indexer /etc/wazuh-indexer/certs/
chmod 500 /etc/wazuh-indexer/certs/
chmod 400 /etc/wazuh-indexer/certs/*
```

2.4 Password Reset After Cert Regeneration

New certs generate new passwords. Indexer still has old internal passwords — unauthorized on curl to localhost:9200. Reset all passwords:

```
/usr/share/wazuh-indexer/plugins/opensearch-security/tools/wazuh-passwords-tool.sh --change-all
```

Save ALL generated passwords to Vaultwarden immediately. Then restart all services in order:

```
systemctl restart wazuh-indexer && systemctl restart wazuh-manager && systemctl restart wazuh-dashboard
```

2.5 IPv6/IPv4 Localhost Issue

```
Error: [ConnectionError]: connect ECONNREFUSED ::1:9200
```

Cause: opensearch_dashboards.yml configured opensearch.hosts as 'localhost' which resolved to ::1 (IPv6). Indexer only listens on 127.0.0.1 (IPv4). Fix:

```
sed -i 's/opensearch.hosts: https:\\\\localhost:9200/opensearch.hosts:
https:\\\\127.0.0.1:9200/' /etc/wazuh-dashboard/opensearch_dashboards.yml

systemctl restart wazuh-dashboard
```

This was the root cause of the dashboard not loading. Always use explicit IP addresses instead of 'localhost' when IPv6/IPv4 ambiguity could be an issue.

3. Vaultwarden Agent Connected

Agent ID 001 — vaultwarden (192.168.10.70) — Active, Wazuh v4.14.6

Verify manager sees agents:

```
/var/ossec/bin/agent_control -l
```

Check agent logs for connection issues:

```
tail -f /var/ossec/logs/ossec.log
```

4. CIS Benchmark Results — Vaultwarden

Policy: CIS Debian Linux 12 Benchmark v1.1.0

Result	Count
Passed	70
Failed	107
Not Applicable	30
Score	39%

4.1 Key Failure Categories

- **Audit logging (33245-33272)**
auditd not installed — no kernel-level audit trail. Critical for Vaultwarden — any file access or command execution is unlogged at the OS level.
- **Password policies (33183-33211)**
libpam-pwquality not installed, no password complexity enforcement, no expiry configured. Critical given Vaultwarden stores all credentials.
- **SSH hardening (33157-33175)**
PermitRootLogin not disabled, MaxAuthTries not configured, no login banner. Reduces brute force protection.
- **Filesystem partitioning (33010-33035)**
/tmp, /var, /var/log not on separate partitions with noexec/nosuid. Harder to fix in LXC — lower priority.
- **Firewall (33126-33151)**
nftables rules incomplete — no default deny policy, loopback not configured.

4.2 Vulnerability Detection

Wazuh automatically scanned installed packages against CVE databases:

Severity	Count
Critical	7
High	52

Fix: apt update && apt upgrade -y on CT102. Running outdated packages on a container holding all credentials is high risk.

5. Complete Upgrade Reference (Future Sessions)

Follow this exact sequence for any future Wazuh upgrade:

1. Take snapshot: pct snapshot 104 pre-wazuh-upgrade-
 2. Upgrade in order: apt install wazuh-indexer -y && apt install wazuh-manager -y && apt install wazuh-dashboard -y
 3. Regenerate certs: rm wazuh-install-files.tar && bash wazuh-install.sh -g && tar -xvf wazuh-install-files.tar
 4. Copy dashboard certs: cp dashboard.pem and dashboard-key.pem to /etc/wazuh-dashboard/certs/
 5. Copy indexer certs: cp node-1.pem (as wazuh-indexer.pem) and node-1-key.pem to /etc/wazuh-indexer/certs/
 6. Copy root-ca.pem to both cert directories
 7. Fix ownership: chown -R wazuh-dashboard:wazuh-dashboard /etc/wazuh-dashboard/certs/ && chown -R wazuh-indexer:wazuh-indexer /etc/wazuh-indexer/certs/
 8. Fix permissions: chmod 500 on dirs, chmod 400 on cert files
 9. Reset passwords: wazuh-passwords-tool.sh --change-all — save ALL to Vaultwarden
 10. Restart in order: wazuh-indexer, wazuh-manager, wazuh-dashboard
 11. Verify indexer: curl -k -u admin: https://localhost:9200/_cluster/health?pretty
 12. Confirm opensearch.hosts uses 127.0.0.1 not localhost in opensearch_dashboards.yml
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6. Next Steps

- Install Wazuh agents on CT103 (Homepage) and pve1 (Proxmox host)
 - Configure OPNsense syslog to forward firewall logs to Wazuh (port 1514)
 - Add wazuh.mystikal.dev subdomain to Caddy
 - Address CIS benchmark failures on Vaultwarden — priority: audit logging, SSH hardening, package updates
 - Update Vaultwarden packages to clear 7 critical CVEs (apt upgrade on CT102)
 - Automate post-upgrade cert fix with Ansible playbook
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